



A Decentralized Cooperative for Secure Communication, Compute and Commerce

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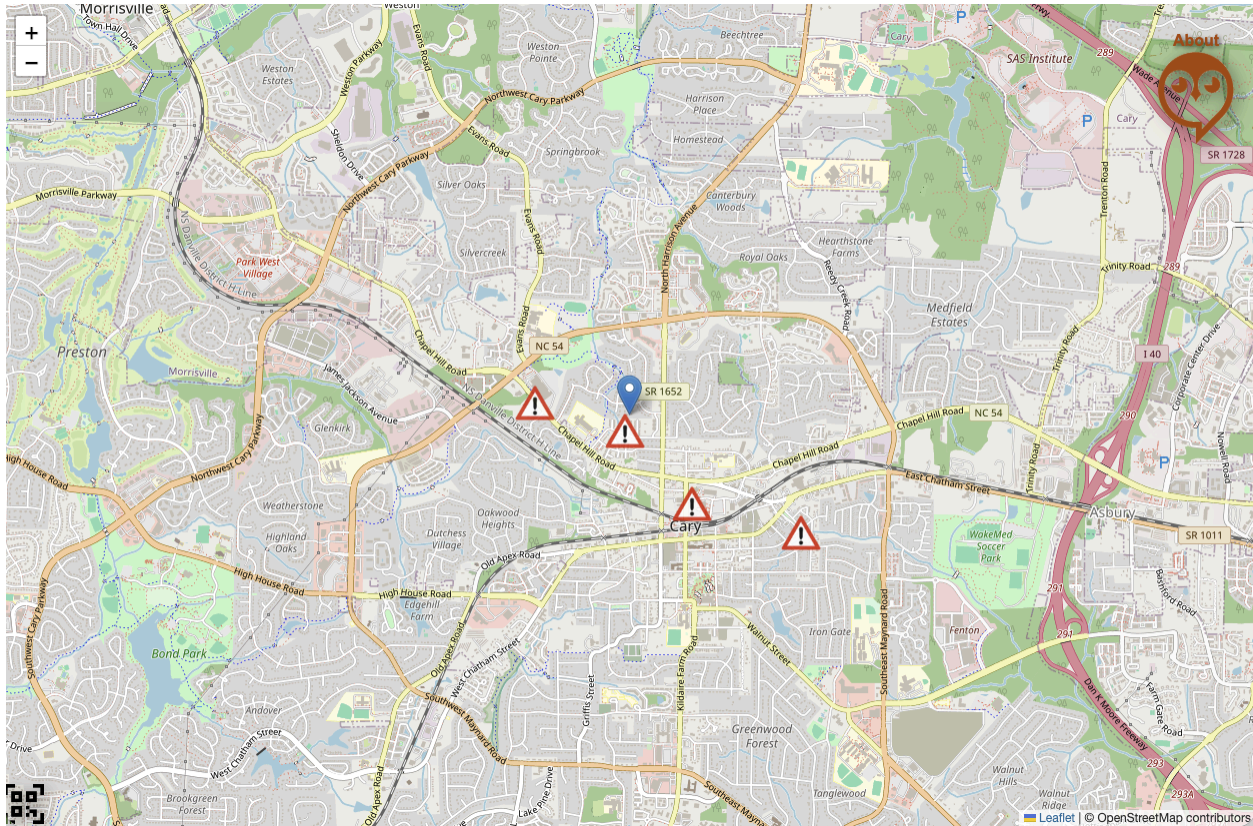
YZ.social (Wise social) is building a totally free and open source, fully secure, peer-to-peer network for a new class of applications. The YZ network has no servers, no central database, no single point of failure. It is a true, fully decentralized network constructed, controlled and owned by its users.

YZ Alert

The first application YZ.social is building on this new network is YZ Alert (Wise Alert) – a way to instantly inform the people in your neighborhood about critical immediate concerns.

You can try an early version of it here. Please don't share this yet:

<https://yz.social>



When launched, YZ Alert displays a map centered on your location. You can tap anywhere on this map to place an alert to other local users of the application. This will highlight the region that the alert is in along with a display of an appropriate icon (fire, accidents, persons of interest).

We are deploying YZ Alert for three reasons. First, it fills a pressing need. Bad actors exist in many communities around the world today; having a platform that alerts members of that community - especially those that are most vulnerable - is essential. Second, this is a proof of the capabilities of a fully decentralized software architecture with unlimited scale. YZ Alert is to YZ what books were to Amazon. We are building an application directly aligned with the mission of the organization. Third is to create the necessary visibility for the YZ.social platform and that mission - creating a powerful platform that provides a safe place for humans to engage with each other.

YZ Network: Secure Peer-to-Peer Network

The YZ network is a pure peer-to-peer ecosystem. There is no central server, no control points. Each user runs as a node in a massively connected and dynamically reconfigurable graph. Communication between nodes can be direct where the nodes are directly connected, or indirect where the messages are routed through the network. The network provides secure connections

not just between the sender and the receiver, but between any two nodes that are forwarding a message to its destination.

Once the network achieves a critical mass of nodes it will be extremely difficult to bring it down. There are no central attack surfaces – each node is independent with redundant edge connections – so even if a large percentage of the nodes are taken offline, the system will continue to function robustly. Even the main entry website, where users initially connect to the system, will be mirrored.

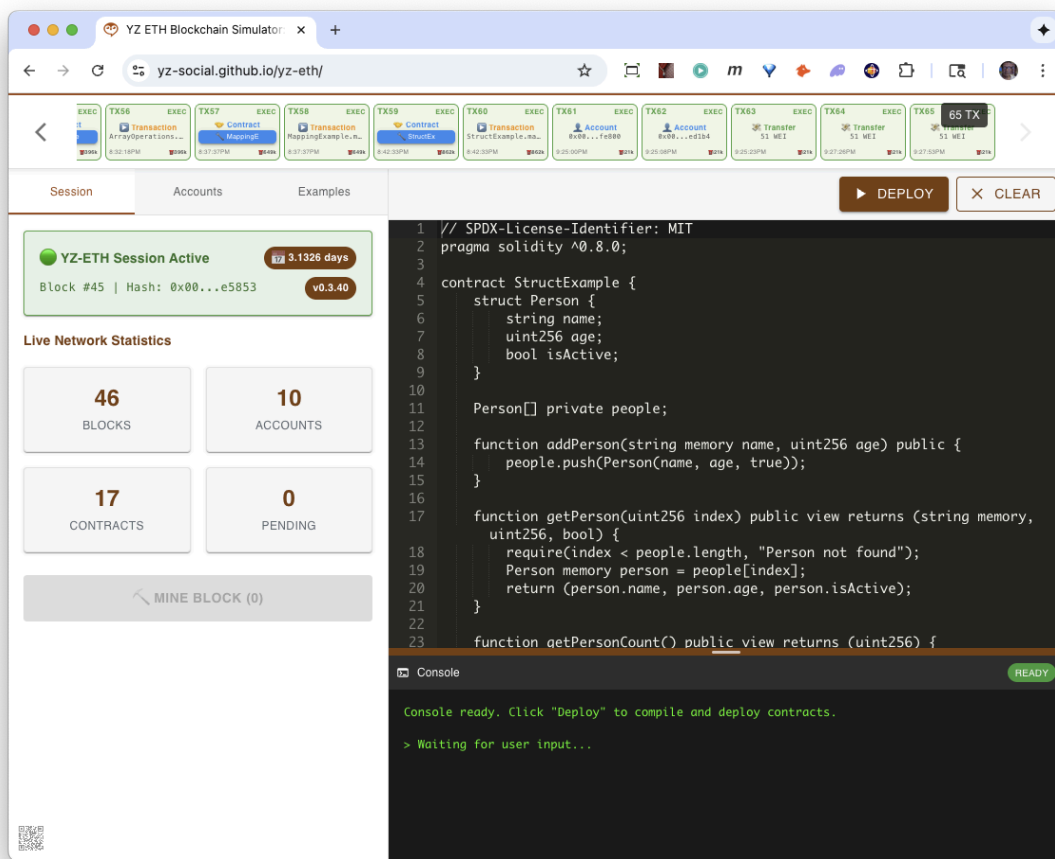
The YZ network is a kind of living thing. Once born, it will never die. Each of its cells may continuously be replaced, but it won't ever fail. Every cell continues to communicate with any other cell, quickly and securely.

The foundation layer of the YZ network is a Kademlia-based distributed hash table (K-DHT), a peer-to-peer storage and messaging system with the additional capability of dynamically constructing application specific topologies. A number of these topologies will be introduced including a local K-DHT that dramatically reduces local geographic communication latency for many applications, and the Concordance Engine, a Croquet descendent, that provides dynamic processes and multi-user capabilities. A node-based publish/subscribe architecture provides a means to broadcast events to any number of nodes as well as to and from virtual dynamic networks with no fixed address such as a Concordance Engine process.

The system design incorporates a decentralized ledger built with the Concordance Engine which provides high speed proof of consensus for transactions. A blockchain-like infrastructure is required to ensure that the community provided compute and communication services are properly compensated and that the network resources are not wasted. The ledger infrastructure is managed as a cooperative, with all participants providing services as required and all participants equally rewarded based upon proof of availability. Among other uses, the ledger is also used to manage the costs of using the system in much the same way that time-sharing systems developed in the 1960s were used.

YZ ETH Studio

As another example, the YZ ETH Studio is a fully functional Ethereum compiler, virtual machine and live blockchain demonstrating Proof of Consensus with the Concordance Engine. It is a multiuser application where smart contracts and transactions can be added by any participant, and the new block is computed by all users. Every transaction within the chain is displayed as a tile in the scrollable list at the top. You can try it out now on your phone or PC at: <https://yz-social.github.io/yz-eth/>



You can create new accounts, transfer tokens between accounts, deploy smart contracts and execute functions within a contract. A new block is minted every 15 seconds. Non-executed transactions appear as brown and executed are green. An identical hash resulting from the minted block is displayed at the top left under the session tab.

This is a simple demonstration of a truly decentralized blockchain application. YZ is designing a scalable, sharded blockchain infrastructure using the Concordance Engine where all members of YZ.social will be participating as miners and users. All participants will be able to support the computation of blockchain transactions, block and other validations, storage and system coordination and will be equally rewarded by providing proof of availability. The actual system will not use the Ethereum virtual machine, as the Solidity language was not designed to operate in a sharded system.

YZ.social: A Cooperative

YZ.social is a cooperative - owned completely by its users. Since the network itself is managing both the ledger and the payments for usage of the system, the participants in the network - its owners - are the ones that are rewarded for this usage.

This document is intended to provide guidance to both the development effort and to those interested in supporting the project. It is certain that the actual deployed system will evolve and diverge from this description, though the functionality and goals of the system will remain constant.